The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$. For each node i, the vector decomposes into six distinct socio-pragmatic and deontic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

where $\boldsymbol{e}_i$ encapsulates the Barna loop splits and Pew macro-advection wavelets, $\boldsymbol{g}_i$ tracks continuous JAX coordinate steps, $\boldsymbol{p}_i$ and $\boldsymbol{r}_i$ pack the level-k PPL choice models and Rational Speech Act distributions, $\boldsymbol{s}_i$ maps the Searlean speech-act flags (X counts as Y in C), and $\boldsymbol{t}_i$ maintains the time-bounded legal compliance trajectories.

3 Homomorphic Multi-Agent Lattices and Fitting Consensus

We formalize the secure distributed multi-agent system by defining a structural identity mapping between the epistemic authority poset and the Functional Zanzibar security perimeter.

3.1 Lattice Homomorphism and Isomorphic Clearance Mappings

Let $\mathcal{A} = \{a_0, a_1, a_2, a_3\}$ represent the set of decentralized neuro-symbolic agents. The architecture unifies the Epistemic Knowledge Lattice $\mathcal{L}_K$ and the Security Flow Lattice $\mathcal{L}_S$ into an identical branching partial-order structure. Instead of a linear sequence, parallel worker agents a_1 and a_2

occupy the same tier coordinate, formalizing a non-linear poset governed by the following strict axiomatic constraints:

$$a_3 \preceq a_2, \quad a_3 \preceq a_1 \tag{2}$$

$$a_2 \preceq a_0, \quad a_1 \preceq a_0 \tag{3}$$

where a_0 (`agent_0_master`) acts as the supreme bounding apex ($\top$) with an epistemic rank of 3, and a_3 (`agent_3_observer`) represents the minimal lower bound ($\bot$) matching `public_base`. The structural identity mapping guarantees that an agent's coordinate position dictates its Zanzibar causal authorization layer natively:

$$\text{Rank}(a_k) \equiv \text{Clearance}(a_k) \tag{4}$$

3.2 Fitting Antimonotonic Logic and MCP Enforcement

Data packet and vocabulary transport across varying clearance domains are executed via the Model Context Protocol (MCP) gateway, which directly enforces Fitting (1992) intuitionistic modal logic semantics. For any two coordinates $w \preceq_T t$, the system enforces antimonotonic perception constraints. The satisfaction relation over local decision forest frames $\mathcal{F}$ yields:

$$M \models d_t(\phi) \implies M \models d_w(\phi) \tag{5}$$

When a consensus conflict is triggered due to a coarse, low-clearance agent a_w being gullible to an un-stabilized path while a sharp, high-clearance agent a_t detects an algebraic boundary anomaly, the MCP gateway applies Clause (d) Intuitionistic Implication Routing and Clause (e) Modal Negation. The low-priority transaction request is instantly dropped, preserving macro-systemic privacy and boundary integrity without black-box VRAM overhead.